Remote Sensing Scene Classification via Second-Order Differentiable Token Transformer Network

Kang Ni, *Member, IEEE*, Qianqian Wu, Sichan Li, Zhizhong Zheng, and Peng Wang, *Senior Member, IEEE*

***Abstract*— The vision transformer has been widely applied in remote sensing image scene classification due to its excellent ability to capture global features. However, remote sensing scene images involves challenges such as scene complexity and small interclass differences. Directly utilizing the global tokens of the transformer for feature learning may increase computational complexity. Therefore, constructing a distinguishable transformer network that adaptively selects tokens can effectively improve the classification performance of remote sensing scene images while considering computational complexity. Based on this, a second-order differentiable token transformer network (SDT^{2}Net) is proposed for considering the efficacy of distinguishable statistical features and nonredundant learnable tokens of remote sensing scene images. A novel transformer block, including an efficient attention block (EAB) and differentiable token compression (DTC) mechanism, is inserted into SDT^{2}Net for acquiring selectable token features of each scene image guided by sparse shift local features and token compression rate learning style. Furthermore, a fast token fusion (FTF) module is developed for acquiring more distinguishable token feature representations. This module utilizes the fast global covariance pooling algorithm to acquire high-order visual tokens and validates the effectiveness of classification tokens and high-order visual tokens for scene classification. Compared with other recent methods, SDT^{2}Net achieves the most advanced performance with comparable floating point operations per second (FLOPs). The code will be available at https://github.com/RSIP-NJUPT/SDT2Net.**

***Index Terms*— Classification token, learnable token, remote sensing, scene classification, vision transformer.**

I. Introduction

REMOTE sensing scene classification is a crucial component of remote sensing image processing and serves as a significant basis for subsequent project development and military operations [1], [2], [3], [4], [5], [6], [7]. Its objective is to categorize scene images into different semantic classes and has been widely applied in areas such as land resource management, urban planning, and natural resource detection [8], [9], [10]. Compared to traditional optical images, remote sensing scene images exhibit characteristics such as scene complexity, intricate textures, and insufficient color information. Consequently, remote sensing scene classification encounters the following challenges.

1) Remote sensing scene images often encompass larger areas, characterized by more intricate distributions of objects and spatial arrangements. Thus, enhancing the effectiveness of feature representation is pivotal in improving the performance of remote sensing image scene classification.
2) Remote sensing scene images exhibit a diverse range of categories, presenting similarities among interclass images and variability within intraclass images. Leveraging the contextual semantic features cross various scenes facilitates a more precise representation of scene images.

Benefiting from the rapid advancement of deep learning theories, methods based on deep learning [11] have become a mainstream approach in current research for remote sensing scene classification [12], [13], [14], [15], [16], [17], [18]. For discriminative feature learning, Xu et al. [19] inserted lie group machine learning approach into the convolutional neural networks (CNNs) for capturing more distinguishing features

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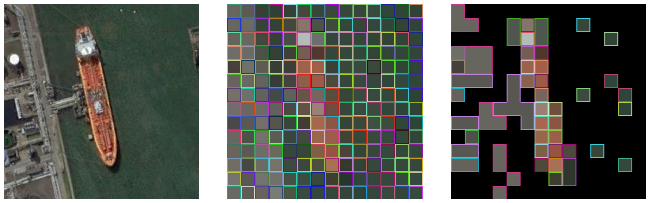
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(a) (b) (c)

Fig. 1. Comparison of token compression before and after. Token compression involves token pruning and merging. Pruned tokens are represented by black and nonboundary markers, while merged tokens are represented by patches with the same internal and boundary colors. (a) Original image of a ship. (b) Feature map containing all 197 tokens. (c) After token compression, there are 35 remaining tokens in the scene image.

of scene imagery. Lv et al. [20] proposed a spatial–channel feature preserving vision transformer (SCViT) network including a lightweight channel attention mechanism on classification token, for acquiring detailed geometric information of the scene. Considering the high-frequency feature learning, Zhang et al. [21] embedded Laplacian high-frequency convolutional block into CNNs for acquiring powerful high-frequency features of remote sensing scene images. It is worth noting that high-order features can effectively characterize the statistical information of images. In recent years, there has also been some research focusing on scene classification of remote sensing images based on high-order statistical features, such as [22], [23], [24]. The approach validates the efficacy of high-order features in the classification of remote sensing scenes. However, most existing methods merely establish high-order statistical measures of local features, lacking comprehensive learning of global high-order features and exhibiting high computational complexity. Hence, the rapid and effective construction of global statistical features for remote sensing scene images is paramount for the practical application of these algorithms.

Due to its multihead self-attention (MHSA) mechanism, the transformer exhibits significant advantages in global feature learning [25], [26]. As a result, it has been widely adopted in remote sensing image scene classification tasks. Chen et al. [27] embedded a lightweight adaptive channel compression module and a hierarchical transformer merging block into transformer architecture for acquiring rich contextual features of scene images. Sha and Li [28] inserted instance mechanism and attention-based multilayer perceptron into the transformer for a better understanding of remote sensing scenes. For exploring contextual information, Tang et al. [29] utilized a multiscale transformer and cross-level attention learning method to obtain global and multilevel discriminative features of scene images. Although the transformer has demonstrated superior performance in scene classification of remote sensing images, the mechanism of global feature learning in the transformer suggests that utilizing all token features to characterize scene features is redundant and may increase model computational complexity. Model compression and pruning are effective methods for reducing model complexity, as illustrated in Fig. 1. Therefore, designing effective token compression and pruning strategies for remote sensing scene classification is crucial for accurately capturing scene features.

In this work, considering the efficacy of leveraging high-order statistical features and nonredundant learnable tokens for effective scene classification in remote sensing imagery, we propose a second-order differentiable token transformer network, named SDT^{2}Net. Our idea aims to fully leverage the advantages of transformer network feature learning by embedding shift operations to enhance its sensitivity to local scene features. In addition, for learnable token selection, we design a differentiable token selection and compression mechanism to adaptively compress and select token features that are more discriminative for different scene images. Finally, considering the complexity of scene images, we effectively describe the statistical information of scene images using high-order features and utilize the fusion vector of classification tokens and high-order visual tokens as the final feature representation. The main contributions are summarized as follows.

1) We design a novel transformer block for scene classification, including an efficient attention block (EAB) and differentiable token compression (DTC) mechanism. While considering local features, this block formulates the token compression rate learning style as an optimization problem embedded within the network for parameter learning. This significantly enhances the model's adaptability for complex scenes.
2) We develop a fast global covariance pooling style with structured normalization via the matrix Taylor polynomial method for exploiting the effectiveness of the classification tokens and visual tokens for remote sensing scene classification. Then, the proposed fast token fusion (FTF) module can be viewed as a novel classification head for scene classification.
3) Extensive experimental results on three remote sensing scene classification benchmarks demonstrate that our SDT^{2}Net achieves competitive scene classification performance with lower floating point operations per second (FLOPs) requirements.

II. Related Work

A. Transformer for Remote Sensing Scene Classification

Remote sensing scene images exhibit intricate spatial structures and encompass diverse land cover types. Therefore, the method used to characterize scene features is of significant importance. Then, there are also many works [30], [31], [32] that attempted to employ the improved transformer for remote sensing scene classification due to its robust capacity for modeling global features. Peng et al. [33] proposed a local–global feature interactive learning architecture for capturing powerful local–global features of scene images. Wang et al. [34] inserted a self-distillation contrastive learning style into the transformer backbone for contextual feature learning. For spatial and frequency feature learning, Yang et al. [35] proposed a transformer-based network that can acquire the global spatial–frequency features between different scales. Simultaneously, there has been a number of research exploring the integration of multiscale contextual information with transformer architectures, such as [26], [36], [37]. The aforementioned approach comprehensively addresses the issue of incorporating contextual information from the scene alongside mitigating the loss of local image features. By employing

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strategies [38], [39], [40], [41] such as CNNs, multiscale analysis, and attention mechanisms, a comprehensive and efficient learning of scene features is achieved. Currently, there are transformer-based approaches employed in multilabel scene classification tasks, such as [42], [43]. However, the majority of these algorithms primarily focus on effectively integrating local features with global contextual features, without considering the discriminative nature of token features and the necessity of all tokens. Consequently, integrating local feature learning into MHSA mechanisms and efficiently enhancing token features while selecting and compressing them can significantly improve the performance of scene classification.

B. Token Compression in Transformer

Token compression aims to reduce the number of input tokens, thereby decreasing the complexity and computational costs of the model while preserving semantic information as much as possible [44], [45], [46], [47], [48]. Token pruning and token merging (ToMe) methods are effective ways of compressing tokens. Bolya et al. [49] proposed a ToMe to combine tokens and instinctively utilize the duplicative elements within the input, making it adaptable to any form of information that contains redundancies. Liang et al. [50] inserted an attention mechanism into MHSA and feed-forward network (FFN) for preserving selected visual tokens. Zeng et al. [51] introduced a token clustering transformer that produces vision tokens with diverse locations, sizes, and shapes for every image through iterative clustering and ToMe mechanisms. These approaches emphasized the creation of standards for distinguishing redundant tokens, additionally, other related algorithms employed a threshold-based approach for token pruning. Although the aforementioned methods utilize either attention mechanisms or threshold-based approaches for token pruning and merging [52], [53], they still lack a DTC method integrated within the transformer framework to facilitate adaptive token pruning and merging. Moreover, considering the complex spatial structures inherent in remote sensing scene images, it is imperative to account for the influence of local features on the adaptive selection and compression of distinguishable tokens. The proposed SDT^{2}Net enables token pruning and ToMe to be adaptively performed in a differentiable manner within a given FLOPs constraint while considering local feature learning of scene images.

III. Methodology

In this section, we first present the overall architecture of the proposed SDT^{2}Net. Then, three core modules are discussed in detail, including EAB, DTC, and FTF modules.

A. Overall Architecture

The overall pipeline of SDT^{2}Net is shown in Fig. 2(a). SDT^{2}Net utilizes DeiT as the backbone, EAB effectively extracts local structural information from images by using shift convolution (Shift-Conv) on the query matrix and key matrix, while maintaining the same computational complexity as 1×1 convolution. In DTC, tokens are pruned and merged sequentially based on the compression rate in each parameter update stage, greatly reducing computational complexity. The block formulates the token compression rate learning style as an optimization problem embedded within the network for parameter learning, achieving the goal of differentiability and learnability. FTF combines class tokens and advanced visual tokens to classify the remote sensing scene images. For the visual token fusion module, a fast matrix Taylor polynomial is employed to acquire the powerful feature statistics via the square root of the covariance matrix.

B. Efficient Attention Block

To train ViT in an end-to-end pipeline, scene images are first split into nonoverlapping patches termed "tokens." These tokens are reshaped into a sequence of flattened fixed-size feature vectors $\mathbf{T}_f \in \mathbb{R}^{N\times C}$ with length N and size C. The trainable position vectors $\mathbf{E}_{\text{pos}}$ are inserted into each token, followed by a classification token $\mathbf{T}_{\text{cls}}$ at the sequence start, which is used for classification. Then, the embedding stage can be formulated as follows:

$$\mathbf{T} = \left[\mathbf{T}_{\text{cls}}; \mathbf{T}_f\right] + \mathbf{E}_{\text{pos}}. \tag{1}$$

The sequences are fed into the ViT encoder block to consider the interactions of each token, including multiple blocks with an MSA and a two-layer MLP. The forward propagation of the ViT block is expressed as follows:

$$\begin{cases} \hat{\mathbf{T}}^{\ell} = \text{MSA}\left(\text{LN}\left(\mathbf{T}^{\ell-1}\right)\right) + \mathbf{T}^{\ell-1} \\ \mathbf{T}^{\ell} = \text{MLP}\left(\text{LN}\left(\hat{\mathbf{T}}^{\ell}\right)\right) + \hat{\mathbf{T}}^{\ell} \end{cases} \tag{2}$$

where $\ell \in [L]$ and L denotes the network depth, $\mathbf{T}^{\ell-1}$ is the output token sequence acquired in block $\ell - 1$, $\mathbf{T}^{\ell}$ is the sequence generated after the MSA mechanism, $\hat{\mathbf{T}}^{\ell}$ is the output sequence of block ℓ, and LN represents the layer normalization (LN). MSA is composed of multiple single-head self-attentions in parallel. The self-attention mechanism models the global interactions of each input vector. First, three trainable weight matrices called queries $\mathbf{W}^Q$, keys $\mathbf{W}^K$, and values $\mathbf{W}^V$ are defined and randomly initialized. The input vectors of the self-attention mechanism are projected into the three matrices to acquire $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$. The query vector is matched with the keys vector to measure the similarity by the dot-product operation $\mathbf{QK}^T$ scaled by $\sqrt{C}$ to achieve normalization and alleviate the vanishing gradient. Then, the attention matrix $\mathbf{A}$ is given by Softmax operation as follows:

$$\mathbf{A} = \text{Softmax}\left(\mathbf{QK}^T/\sqrt{C}\right). \tag{3}$$

The proposed EAB effectively captures local structural features from scene images by utilizing the shift convolution method to acquire the query matrix and key matrix. In the shift convolution module (SCM) shown in Fig. 2(b), we use the Shift-Conv method, which maintains the same computational complexity as the 1×1 convolution. Previous studies have typically used multilayer perceptrons or two cascaded 1×1 convolutions to obtain local features. But these convolutions only have a 1×1 receptive field, and each convolution kernel can only perceive its local region, while Shift-Conv introduces translation operations, allowing the convolution kernel to perceive information from adjacent

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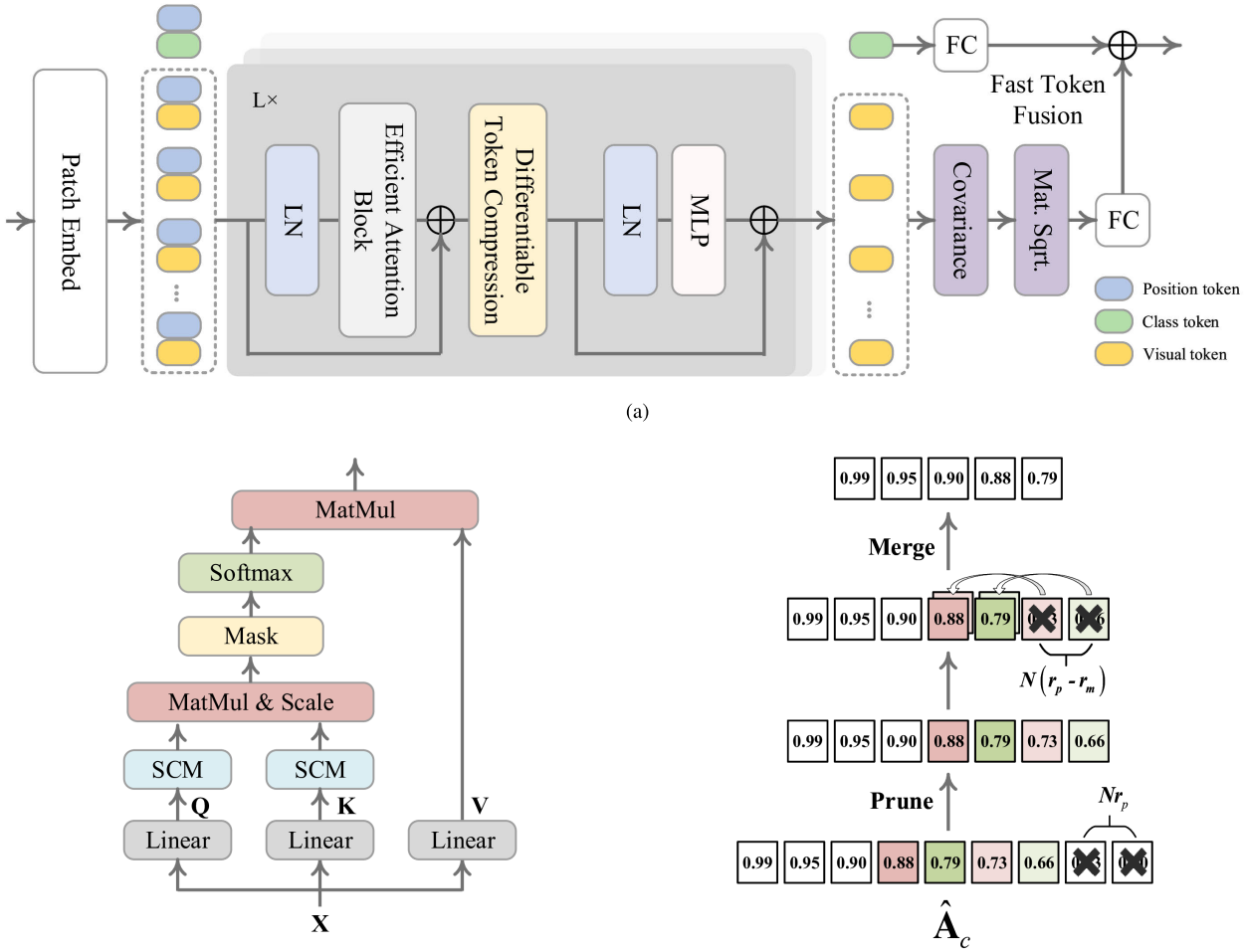


Fig. 2. Illustration of SDT^{2}Net. (a) Overall pipeline of SDT^{2}Net. (b) Architecture of EAB. (c) Illustration of DTC.

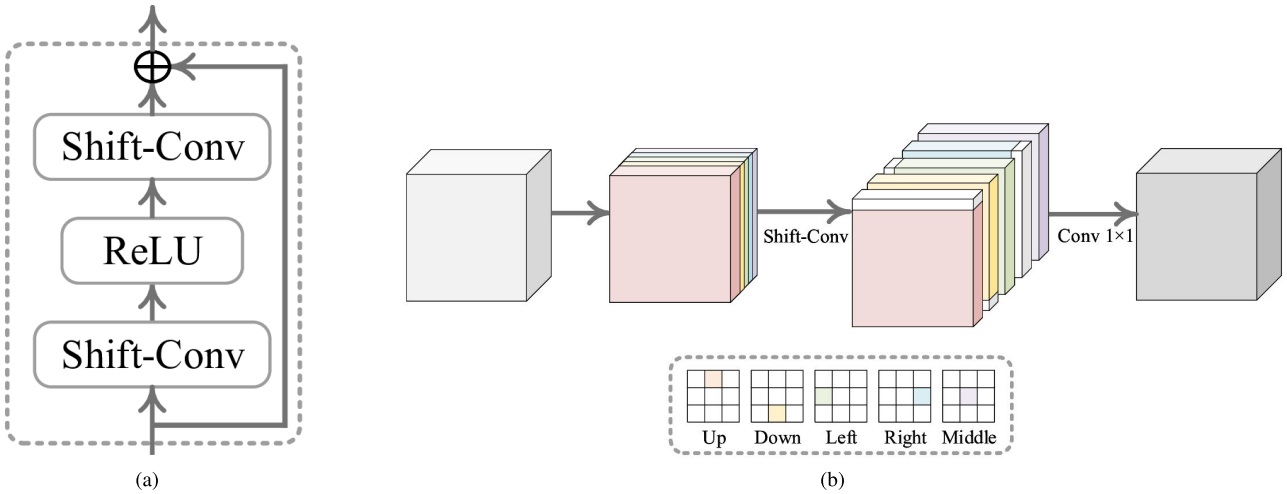


Fig. 3. Overall pipeline of SCM. (a) Illustration of SCM. (b) Illustration of Shift-Conv.

pixels. In order to expand the receptive field for more effective local feature extraction, we introduce two Shift-Convs and a simple ReLU activation function is employed between them, as shown in Fig. 3(a). Shift-Conv consists of a set of shift operations and a 1×1 convolution, as shown in Fig. 3(b). The input features are divided into five groups, performing translation operations in different directions, including up, down, left, and right, and keeping them unchanged in five spatial dimensions. In this way, the features of each group can obtain information about its surrounding pixels. Next, a

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1×1 convolution operation is performed on the translated features of each group to fuse and learn local features. Then, the attention mechanism is now computed on $\mathbf{Q}$ and $\mathbf{K}$ with the SCM operation, as follows:

$$\mathbf{G} = \text{Softmax}\left(\text{SCM}(\mathbf{Q})\text{SCM}(\mathbf{K})^T/\sqrt{C}\right). \tag{4}$$

In order to facilitate subsequent description, we mark $\mathbf{Q}$ and $\mathbf{K}$ after SCM operation as $\hat{\mathbf{Q}}$ and $\hat{\mathbf{K}}$.

The design of this Shift-Conv aims to provide a larger receptive field without introducing additional learnable parameters, maintaining almost the same computational complexity as 1×1 convolution. Therefore, it is possible to improve the performance of the model while maintaining computational efficiency. Through this method, we can more fully capture the details and local structural information of scene images.

C. Differentiable Token Compression

Token pruning and merging modules are performed through different compression rates in DTC, and a single variable r is utilized to represent pruning and merging compression rates. To achieve differentiable compression rates while maintaining a constant number of tokens and minimizing interactions among pruned tokens, compression rates are redefined as a dynamic style of initial compression rates, enhancing computation efficiency

$$r = \sum_{j=1}^{N} p_j D_j \tag{5}$$

where $\mathbf{D} = \{D_1, D_2, \ldots, D_N\}$ symbolizes a set of candidate compression rates, each rate $D_j = (j-1)/N$ signifies the elimination of the least significant $j-1$ tokens. Here, p_j represents the learnable probability assigned to each rate within D_j, and $\sum_{j=1}^{N} p_j = 1$. Then, the probability of compressing the jth less significant token can be calculated as

$$\pi_j = \begin{cases} \sum_{j=1}^{N-1} p_{j+1}, & j \neq N \\ 0, & j = N. \end{cases} \tag{6}$$

From (6), one can observed that $\pi_j \geq \pi_{j+1}$ indicates a higher compression probability for less related tokens. Notably, $\pi_N = 0$ ensures the preservation of the most related token, preventing the compression of all tokens. For consistency in the training and inference stages, π_j is transformed into a binary mask as given by

$$P_j = \begin{cases} 1, & \pi_{N+1-j} < r \\ 0, & \pi_{N+1-j} \geq r \end{cases} \tag{7}$$

$$P_j^{\ell} = P_j^{\ell-1} \cdot P_j^{p} \cdot P_j^{m} \tag{8}$$

where $P_j = 1$ indicates that the jth token is preserved. Within each ViT block, two sets of dynamic candidate rates are deployed to acquire both the pruning and merging compression rates, resulting in two token masks: pruning mask P^p and merging mask P^m. It's important to note that tokens eliminated in the preceding block are also compressed in the current one, leading to the establishment of the final decision mask P_j^{ℓ} in the ℓth block as outlined in (8).

Due to the unstructured information of decision mask P_j and different counts of tokens across samples, directly removing tokens marked by $P_j = 0$ leads to inconsistent token numbers in a batch. To maintain uniform token counts while minimizing pruned token interactions, we employ attention masking M_{ij}, inspired by DynamicViT [54], using P_j removed the subscript ℓ for simplification. This approach effectively negates the influence of discarded tokens, ensuring efficient computation without impacting the self-attention mechanism

$$M_{ij} = \begin{cases} P_j, & i \neq j \\ 1, & i = j \end{cases}, \quad 1 \leq i, \ j \leq N \tag{9}$$

$$\tilde{\mathbf{A}}_{ij} = \frac{\exp(\mathbf{G}_{ij})\mathbf{M}_{ij}}{\sum_{k=1}^{N} \exp(\mathbf{G}_{ik})\mathbf{M}_{ik}}, \quad 1 \leq i, \ j \leq N. \tag{10}$$

Equation (10) formulates the adjusted attention matrix $\tilde{\mathbf{A}}_{ij}$, maintaining the original attention matrix's effect with a consistent $N \times N$ size for the preserved tokens during the training stage.

In the network backpropagation, the straight-through-estimator (STE) [55] is utilized to acquire the gradient of (9). Therefore, we can use chain rules to calculate the gradient of loss $\mathcal{L}$ relative to p_j

$$\frac{\partial \mathcal{L}}{\partial p_j} = \sum_{j=1}^{N} \frac{\partial \mathcal{L}}{\partial P_j} \frac{\partial P_j}{\partial \pi_j} \frac{\partial \pi_j}{\partial p_j} \approx \sum_{j=1}^{N} \frac{\partial \mathcal{L}}{\partial P_j} \frac{\partial \pi_j}{\partial p_j}. \tag{11}$$

Given that p_j is differentiable via (11), we can optimize the compression rate r with gradient back-propagation as shown in (5).

As illustrated in Fig. 2(a), the attention masking is performed on the Mask module in EAB. After updating the compression rate in the training stage, DTC compresses unimportant tokens based on the pruning rate r_p and merging rate r_m. It is worth noting that, the superscript ℓ of the compression rate is dropped to simplify the notation in Fig. 2(c). Here, we employ the class attention $\hat{\mathbf{A}}_c$ which measures how much each image token contributes to the class token, as the importance metric following EViT [50]

$$\hat{\mathbf{A}}_c = \text{Softmax}\left(\hat{\mathbf{Q}}_c \hat{\mathbf{K}}^T/\sqrt{C}\right) \tag{12}$$

where $\hat{\mathbf{Q}}_c$ denotes the class token of query vector $\hat{\mathbf{Q}}$, and $\hat{\mathbf{K}}$ denotes the key vector $\mathbf{K}$ after SCM operation. As shown in Fig. 2(c), we first rank tokens by their importance, with higher class attention signaling greater significance for the final output [50]. The closer one represents the more important. Then, the less important tokens, often related to irrelevant backgrounds, are pruned based on the established token pruning style. In the ℓth transformer block, Nr_p less important tokens are removed. Finally, the similarity between the remaining and $N(r_m - r_p)$ less important tokens is given by cosine similarity, and similar token pairs are merged via average operation. The token compression style can be summarized as follows:

$$\hat{\mathbf{T}}_c^{\ell} = f_c\left(\hat{\mathbf{T}}^{\ell}, r_p, r_m\right) \tag{13}$$

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TABLE I
DESCRIPTION OF DATASETS NWPU-RESISC45, AID, AND UC MERCED

	NWPU-RESISC45	AID	UC Merced
Total images	31500	10000	2100
Total classes	45	30	21
Images per class	700	220-420	100
Image size	256×256	600×600	256×256
Spatial resolution (m)	0.2-3	0.5-8	0.3
Scene classes	Airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland.	Airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct.	Agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tank, and tennis court.

where f_c represents the token compression method, $\hat{\mathbf{T}}^{\ell}$ is the output of attention module in (3), and $\hat{\mathbf{T}}_c^{\ell}$ is the output of $\hat{\mathbf{T}}^{\ell}$ after DTC module. Through the above pruning-merging approach, SDT^{2}Net dynamically selects the optimal number of tokens for pruning and merging in each block using a learnable compression rate, effectively pruning and merging tokens in an adaptive learning style.

D. FTF Module

To enhance the powerful semantic information in visual tokens, which traditional transformers may overlook by relying solely on class tokens for prediction, an FTF module is utilized. This module leverages both class and advanced visual tokens for classification, aiming to enrich the usage of visual data and boost classification performance. Let $\mathbf{z}_0$ indicate the classification token while $\mathbf{Z} = [\mathbf{z}_1; \mathbf{z}_2; \ldots; \mathbf{z}_N] \in \mathbb{R}^{N \times C}$ indicates visual tokens. The combined class prediction y is as follows:

$$y = \text{Softmax}\left(\text{FC}(\mathbf{z}_0) + \text{FC}\left(\left(\mathbf{Z}\mathbf{Z}^T\right)^{\frac{1}{2}}\right)\right) \tag{14}$$

where FC is a fully connected layer. As shown in Fig. 2(a), the covariance square roots of the visual tokens are computed for classification [56].To further reduce computational complexity, a fast matrix Taylor polynomial [57] is given to calculate the square root of covariance $\mathbf{Z}\mathbf{Z}^T$ as follows:

$$\left(\mathbf{Z}\mathbf{Z}^T\right)^{\frac{1}{2}} = \mathbf{I} - \sum_{m=1}^{\infty} \left| \binom{\frac{1}{2}}{m} \right| (\mathbf{I} - \mathbf{Z}\mathbf{Z}^T)^m \tag{15}$$

where $\mathbf{I}$ is an identity matrix, $\binom{(1/2)}{m}$ denotes the binomial coefficients that involve fractions calculated as $(1/2)!/m!((1/2)-m)!$. By substituting $\mathbf{Z}\mathbf{Z}^T$ with $\mathbf{B}$ and ensuring that the norm $\|(\mathbf{I}-\mathbf{B})\|_p$ is less than 1, we maintain the convergence of the power series. $\|\cdot\|_p$ denotes any vector-induced matrix norms, which is achieved by prenormalizing $\mathbf{B}$ with its Frobenius norm $\|\mathbf{B}\|_{\text{F}}$, and $\|\mathbf{I} - (\mathbf{B}/\|\mathbf{B}\|_{\text{F}})\|_p < 1$. The square root of $\mathbf{B}$ is then adjusted postcalculation by the factor $\sqrt{\|\mathbf{B}\|_{\text{F}}}$. Then, (15) can be rewritten as follows:

$$\mathbf{B}^{\frac{1}{2}} = \sqrt{\|\mathbf{B}\|_{\text{F}}} \cdot \left(\mathbf{I} - \sum_{m=1}^{\infty} \left| \binom{\frac{1}{2}}{m} \right| \left(\mathbf{I} - \frac{\mathbf{B}}{\|\mathbf{B}\|_{\text{F}}}\right)^m\right). \tag{16}$$

Cutting off the series at level m provides the matrix Taylor polynomial approximation for the matrix square root. To compute the matrix Taylor polynomial of level m, $m-1$ matrix multiplications are required in FTF.

IV. EXPERIMENTS

In this section, we provide a detailed introduction to the dataset, implementation details, and experimental results to evaluate the performance of SDT^{2}Net.

A. Datasets and Implementation Details

To demonstrate the performance of SDT^{2}Net, validation evaluations are carefully conducted on the following three remote sensing scene classification datasets: NWPU-RESISC45 [58], AID [59], and UC Merced [60]. The details of these datasets are provided in Table I. For each dataset, the samples are randomly divided into training data and testing data.

The SDT^{2}Net is optimized with AdamW optimizer. We trained SDT^{2}Net for 100 epochs with a batch size of 64 with NVIDIA GeForce RTX 24G 3090 GPU based on the PyTorch framework. The weight decay and learning rate were set as 0.05 and 0.01, respectively. The learning rate is scaled by a cosine decay learning rate scheduler.

B. Scene Classification

To quantitatively verify the scene classification performance of SDT^{2}Net, we employ the overall accuracy (OA) metric, which is the accuracy of correctly predicted samples within the total test set. Table II presents the quantitative evaluation, detailing the average OA and its standard deviation.

As shown in Table II, for each of the three datasets, the scene images can be well distinguished from each other and reasonably identified as their corresponding categories

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TABLE II
COMPARISON RESULTS (%) OF THE PROPOSED METHOD AND THE STATE-OF-THE-ART METHOD ON THE DATASETS

Model	Publication (Years)	NWPU-RESISC45		AID		UC Merced	
		10%	20%	20%	50%	50%	80%
CSCANet	GRSL2023	91.27	93.27	94.67	96.83	-	99.76
LHNet	TGRS2022	89.89 ± 0.15	92.53 ± 0.13	93.30 ± 0.10	97.81 ± 0.13	-	99.22 ± 0.11
ViT	TGRS2022	91.59 ± 0.19	93.90 ± 0.20	94.90 ± 0.29	96.49 ± 0.18	98.75 ± 0.21	99.29 ± 0.34
Swin-T	ICCV2021	90.84 ± 0.09	93.18 ± 0.15	94.56 ± 0.14	96.92 ± 0.12	-	99.46 ± 0.11
T2T-ViT-14	ICCV2021	91.32 ± 0.15	93.89 ± 0.12	95.05 ± 0.21	96.99 ± 0.20	98.39 ± 0.63	99.10 ± 0.10
SCViT	TGRS2022	92.72 ± 0.04	94.66 ± 0.10	95.56 ± 0.17	96.98 ± 0.16	98.90 ± 0.19	99.57 ± 0.31
PVTv2-B2	CVM2022	90.62 ± 0.16	93.55 ± 0.29	94.42 ± 0.35	96.40 ± 0.19	98.34 ± 0.11	98.96 ± 0.37
EMTCAL	TGRS2022	91.63 ± 0.19	93.65 ± 0.12	94.69 ± 0.14	96.41 ± 0.23	98.67 ± 0.16	99.57 ± 0.28
C^2-CapsViT	GRSL2022	93.32 ± 0.05	95.28 ± 0.08	96.05 ± 0.11	97.57 ± 0.15	99.33 ± 0.13	99.76 ± 0.12
SF-MSFormer	TGRS2023	-	-	-	98.72 ± 0.31	-	99.72 ± 0.12
EViT	ICLR2022	92.49 ± 0.08	94.46 ± 0.13	94.40 ± 0.29	96.72 ± 0.14	98.71 ± 0.12	99.20 ± 0.05
ToMe	ICLR2023	92.56 ± 0.10	94.65 ± 0.16	95.19 ± 0.09	97.17 ± 0.12	98.54 ± 0.10	99.20 ± 0.07
ViT_CL	JSTARS2023	92.85	94.69	95.60	97.42	99.14	99.76
PSCLI-TF	GRSL2024	92.92	94.86	96.28	97.52	98.70	99.62 ± 0.12
DeiT	ICML2021	93.10 ± 0.07	95.30 ± 0.07	95.50 ± 0.10	97.50 ± 0.07	98.60 ± 0.08	99.29 ± 0.12
SDT^{2}Net	**Ours**	**93.85 ± 0.02**	**95.63 ± 0.01**	**96.36 ± 0.04**	**97.96 ± 0.03**	**99.37 ± 0.05**	**99.89 ± 0.01**

with relatively low misclassification. In comparison, SDT^{2}Net achieves the highest classification accuracy on the UC Merced dataset, while its performance on the NWPU-RESSC45 dataset is relatively low. This is attributed to the larger number of scene categories and more complex scene conditions present on the NWPU-RESSC45 dataset. However, the observed accuracies across various datasets are notably high, with OAs surpassing 93.85% on all experimental scenarios. Specifically, within the large-scale NWPU-RESISC45 dataset, the OAs reach 93.85% and 95.63% for training sample ratios of 10% and 20%, respectively. On the AID dataset, which is of medium scale, OAs are recorded at 96.36% with a 20% training ratio and 97.96% with a 50% training ratio. With regard to the small-scale UC Merced dataset, SDT^{2}Net achieves an OA of 99.37% with 50% training samples and an OA of 99.89% with 80% training samples. The primary contributors to classification inaccuracies are the notable similarities among interclass images, e.g., "palace" and "church" categories within the NWPU-RESISC45 dataset. It is noted that the classification accuracy experiences a marked improvement with more training samples across all datasets, particularly on the NWPU-RESISC45 dataset. This indicates that more training data and a powerful feature learning style could enhance the model's capability to discriminate and its feature representation, which is pivotal in remote sensing scene classification.

C. Comparative Study

To further evaluate the effectiveness of SDT^{2}Net, detailed comparative evaluations and analyses were carried out against numbers of related networks, such as, contextual spatial–channel attention network (CSCANet) [61], Laplacian high-frequency network (LHNet) [21], ViT [29], Swin-T [62], tokens-to-token vision transformer (T2T-ViT-14) [63], SCViT [20], improved pyramid vision transformer (PVTv2-B2) [64], efficient multiscale transformer and cross-level attention learning (EMTCAL) [29], cross-context and cross-scale capsule (C^2-CapsViT) [36], an explainable spatial–frequency multiscale transformer based on ResNet34 (SF-MSFormer) [35], token pruning (EViT) [50], ToMe [49], vision transformer with contrastive learning (ViT_CL) [32], position-sensitive cross-layer interactive transformer model (PSCLI-TF) [32], and DeiT [65]. As our proposed SDT^{2}Net mainly focuses on transformers, then the compared methods are based on ViT, except for CSCANet and LHNet. All the models are trained on three datasets. The comparison results of SDT^{2}Net and the state-of-the-art methods on different datasets are reported in Table II.

While the training rate of the NWPU-RESISC45 dataset is set to 10%, our proposed SDT^{2}Net can achieve the highest accuracy in CNNs-based methods, which is 2.58% and 3.96% higher than CSCANet and LHNet, respectively. However, the enhanced performance of the model based on the ViT method effectively demonstrates the competitive ability of global feature learning, especially C^2-CapsViT and ViT_CL with OA as high as 93.32% and 92.85%, respectively. And our SDT^{2}Net is still the best method, with an OA 0.53% higher than C^2-CapsViT and 1.00% higher than ViT_CL. While the training rate is set to 20%, similar observations can be observed. Our OA of SDT^{2}Net is 0.35% higher than C^2-CapsViT and 0.77% higher than PSCLI-TF. The experimental results indicate that our SDT^{2}Net achieves the best scene classification performance.

For the AID dataset with 20% training ratio, our proposed SDT^{2}Net achieves the highest accuracy in both CNNs-based and ViT-based methods, which is 0.31% higher than the second highest C^2-CapsViT. While the training rate is set to 50%, CNNs-based network, LHNet, shows higher accuracy, only 0.15% lower than SDT^{2}Net. For the ViT-based methods, SF-MSFormer exhibits extremely high accuracy, with an OA of 98.72%. This is due to its introduction of a multiscale transformer branch in both spatial and frequency domains, paying much attention to the global multiscale representation of spatial–frequency features. Nevertheless, it should be noted that the standard deviation of SF-MSFormer is as high as 0.31, which is 0.28 higher than the proposed SDT^{2}Net, indicating some fluctuation in the experimental results.

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For the UC Merced dataset with a training rate set to 50%, SDT^{2}Net achieves the highest classification scores in both CNNs-based and ViT-based methods, 0.04% higher than the second highest C^2-CapsViT. While the training ratio is set to 80%, CSCANet based on CNNs, C^2-CapsViT based on ViT, and ViT-CL all have an OA of 99.76%, demonstrating excellent performance. But they are all 0.13% lower than our proposed SDT^{2}Net. Due to the small size and limited variety of scenes on the UC Merced dataset, the majority of methods have an OA higher than 99%, and the performance gap between them is not as large as that of NWPU45-RESISC45 and AID datasets. These experimental results also indicate that SDT^{2}Net always achieves the highest classification scores.

Specifically, our proposed SDT^{2}Net exhibits significant improvements. Compared with the backbone network DeiT, SDT^{2}Net improves OA by 0.75%, 0.86%, and 0.77% on the NWPU-RESSC45 dataset with a training ratio of 10%, AID dataset with a training ratio of 20%, and UC Merced dataset with a training ratio of 50%, respectively. For a small training ratio, the accuracy of SDT^{2}Net is 1.36%, 1.96%, and 0.66% higher than EViT, which is used for the token pruning approach, and 1.29%, 1.17%, and 0.17% higher than ToMe method based on ToMe strategy. For large training ratio, SDT^{2}Net is 1.17%, 1.24%, and 0.69% higher than EViT, and 0.98%, 0.79%, and 0.69% higher than ToMe, respectively. The proposed SDT^{2}Net combines the advantages of both EViT and ToMe, and it performs token pruning and merging in sequence during the token compression stage, effectively improving classification accuracy. Simultaneously, SDT^{2}Net can classify remote sensing images efficiently using fewer tokens under the computation constraint.

Moreover, as shown in Table II, the classification accuracy of SDT^{2}Net on each dataset surpasses that of other comparison algorithms. Therefore, SDT^{2}Net has played a considerable competitive role in remote sensing image scene classification.

D. Parameter Analysis

1) Effectiveness of Shift-Conv: In the EAB module, SCM aims to capture localized features effectively. To validate its efficacy, the SCM function is replaced with 1×1 and 3×3 convolutional layers under low training ratios across three datasets.

As shown in Table III, it is obvious that the classification performance obtained by 1×1 convolution is much better than that of 3×3 convolution on three datasets. For example, while the FLOPs constraint is 2.3G, the classification accuracy using 1×1 convolution is higher than that of 3×3 convolution by 0.29%, 1.28%, and 1.52%, respectively. Under the same FLOPs constraint, the model performance using SCM module is far better than that of the conventional convolution layer on each dataset, and the average training throughput equals that of a 1×1 convolution. In addition, it's noticeable that halving the FLOPs constraint results in a roughly 2000 images per second improvement in the average throughput across all methods. This demonstrates that as the number of tokens generated by the network decreases, the speed of model inference increases. Based on the aforementioned experiments and analysis, SDT^{2}Net employing the SCM module demonstrates superior model performance while reducing the calculation costs.

TABLE III

VALIDITY VERIFICATION OF SCM MODULE ON NWPU45, AID, AND UCM DATASETS. THE NWPU-RESISC45 DATASET IS REFERRED TO AS NWPU45, AND UC MERCED IS REFERRED TO AS UCM. THROUGHPUT IS REFERRED TO AS TP, AND THE UNIT IS THE NUMBER OF IMAGES INFERRED PER SECOND (imgs/s)

Dataset	Method	OA (%)	FLOPs(G)	TP($imgs/s$)
NWPU45	Conv1×1	92.77 ± 0.03	2.3	6213
	Conv3×3	92.48 ± 0.17	2.3	6036
	SCM	93.39 ± 0.12	2.3	6296
	Conv1×1	93.65 ± 0.01	4.5	4254
	Conv3×3	93.10 ± 0.14	4.5	4115
	SCM	**93.85 ± 0.02**	**4.5**	**4290**
AID	Conv1×1	95.56 ± 0.14	2.3	6151
	Conv3×3	94.28 ± 0.19	2.3	5907
	SCM	95.69 ± 0.06	2.3	6178
	Conv1×1	96.03 ± 0.01	4.5	4215
	Conv3×3	95.81 ± 0.15	4.5	4097
	SCM	**96.36 ± 0.04**	**4.5**	**4198**
UCM	Conv1×1	98.57 ± 0.09	2.3	6186
	Conv3×3	97.05 ± 0.25	2.3	5953
	SCM	98.67 ± 0.11	2.3	6194
	Conv1×1	99.05 ± 0.01	4.5	4222
	Conv3×3	98.86 ± 0.14	4.5	4142
	SCM	**99.37 ± 0.05**	**4.5**	**4205**

TABLE IV

PERFORMANCE COMPARISON OF DIFFERENT TOKEN FUSION STRATEGIES ON NWPU-RESISC45 DATASET

DTC	Token Fusion Strategy	Training Ratio	
		10%	20%
without DTC	only Cls	93.37 ± 0.10	95.40 ± 0.04
	Cls + Vis	93.48 ± 0.08	95.50 ± 0.05
	Cls *concat* Vis	93.36 ± 0.05	95.37 ± 0.14
	our FTF	93.71 ± 0.12	95.59 ± 0.06
with DTC	only Cls	93.45 ± 0.03	95.53 ± 0.02
	Cls + Vis	93.29 ± 0.02	95.47 ± 0.01
	Cls *concat* Vis	93.39 ± 0.04	95.50 ± 0.03
	our FTF	**93.85 ± 0.02**	**95.63 ± 0.01**

2) Analysis of Token Fusion Strategy: Due to the abundant information contained in advanced visual tokens, ignoring them would result in a decrease in classification accuracy. Therefore, our proposed FTF module develops a structured normalized fast global covariance pooling style using the matrix Taylor polynomial method to leverage the effectiveness of classification and visual tokens as the final classifier. To further analyze the contribution of FTF, we conduct experiments on the NWPU-RESISC45 dataset with training ratios of 10% and 20%, respectively, with/without the DTC module, as shown in Table IV.

In the table, "only Cls" represents using only class tokens. "Cls + Vis" represents the aggregated visual tokens after global average pooling and class tokens are, respectively, connected to the FC layer, and then summed before being fed to the softmax classifier. "Cls concat Vis" represents the aggregated visual tokens through fast global covariance pooling and class tokens are concatenated, followed by the

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TABLE V
COMPARATIVE STUDY OF TOKEN COMPRESSION METHODS ON 10% TRAINING RATIO OF NWPU-RESISC45 DATASET

Model	Method	FLOPs(G)	TP($imgs/s$)	OA (%)
DeiT-S	Baseline	4.6	2655	93.10 ± 0.07
	EViT	3.0	4204	91.70 ± 0.22
	ToMe	2.9	4032	92.35 ± 0.14
	SDT^{2}Net	**2.9**	**5194**	**93.42 ± 0.09**
	EViT	4.6	2638	92.49 ± 0.08
	ToMe	4.6	2654	92.56 ± 0.10
	SDT^{2}Net	**4.5**	**4290**	**93.85 ± 0.02**

FC layer and a softmax classifier. "our FTF" represents the token fusion strategy used by SDT^{2}Net, where the aggregated visual tokens that have been quickly pooled through global covariance and the class tokens are, respectively, connected to FC layer and then summed before being fed to the softmax classifier. As illustrated in Table IV, for methods without token compression, the classification accuracy of "only Cls" and "Cls concat Vis" have similar performance, reaching 93.37% and 93.36%, respectively, on the NWPU45-RESISC45 dataset with a training ratio of 10%. At a training ratio of 20%, it is 95.40% and 95.37%, respectively. The proposed FTF token fusion strategy performs the best, with accuracy rates of 93.71% and 95.59%, respectively, which are 0.23% and 0.09% higher than the second highest "Cls + Vis." For the methods using DTC, "Cls + Vis" appears less competitive, with accuracy rates of 93.29% and 95.47%, respectively, under two training ratios, which are 0.56% and 0.16% lower than the results obtained using FTF. For the methods of "only Cls" and "Cls concat Vis," the accuracy of the two methods is still relatively similar under the two training ratios. However, the former is superior, with the accuracy of "only Cls" being 0.06% and 0.03% higher than that of "Cls concat Vis," respectively. The SDT^{2}Net with the FTF module once again achieved the best performance, outperforming "only Cls" by 0.40% and 0.10% in two training ratios.

Overall, SDT^{2}Net captures the statistical characteristics of scene images through high-order features and leverages the fusion vector derived from classification tokens and high-order visual tokens as the final feature representation. The experimental results showcase the efficacy of employing high-order statistical features and learnable tokens for classifying scenes in remote sensing images.

3) Analysis of Existing Token Compression Methods: In order to fully verify the effectiveness of the proposed differential token compression method, we further analyze the time cost of existing token compression methods and SDT^{2}Net. The comparative results of token pruned EViT, token merged ToMe and SDT^{2}Net on NWPU-RESISC45 dataset with a training ratio of 10% are shown in Table V.

For EViT, the FLOPs obtained by the default pruning rate are 3.0G. Compared to DeiT, FLOPs drop by 1.6G, while the model throughput nearly doubles. However, the OA of EViT drops significantly at 91.70%. Moreover, one can obtain similar FLOPs and throughput to EViT by default settings, but ToMe's classification accuracy is 0.65% higher than that of EViT. It is worth noting that FLOPs are limited to around 2.9G, thereby achieving the highest classification efficiency and accuracy for SDT^{2}Net. Especially, SDT^{2}Net is 0.32% and 1.07% higher than baseline and ToMe, respectively. While the FLOPs constraint is increased to around 4.6G of baseline, it can be observed that the classification accuracy of EViT and ToMe is improved by 0.79% and 0.21%, respectively. Due to the growth of FLOPs, the number of images classified per second drops to about 2600. But SDT^{2}Net still acquires the highest model throughput, which is close to EViT while FLOPs are 3.0G. SDT^{2}Net achieves the highest classification scores of 93.85%, which is 1.36% and 1.29% higher than EViT and ToMe methods with similar calculation parameters.

In summary, the DTC method has outstanding model efficiency and performance compared with existing methods under similar FLOPs constraint. Even when facing stringent computational complexity requirements, SDT^{2}Net shows less time cost and better performance, effectively verifying the effectiveness and superiority of the proposed DTC method.

4) Analysis of Model Complexity: To further explore the superiority of the proposed model, we evaluate the performance of SDT^{2}Net under different FLOPs settings. The optimization objective of the token compression method is to minimize the classification loss with the target of FLOPs by leveraging the token pruning rate and merging rate. Equation (11) guides the update of p_j in a differentiable manner to determine the optimal compression rates, which can be directly applied to existing models.

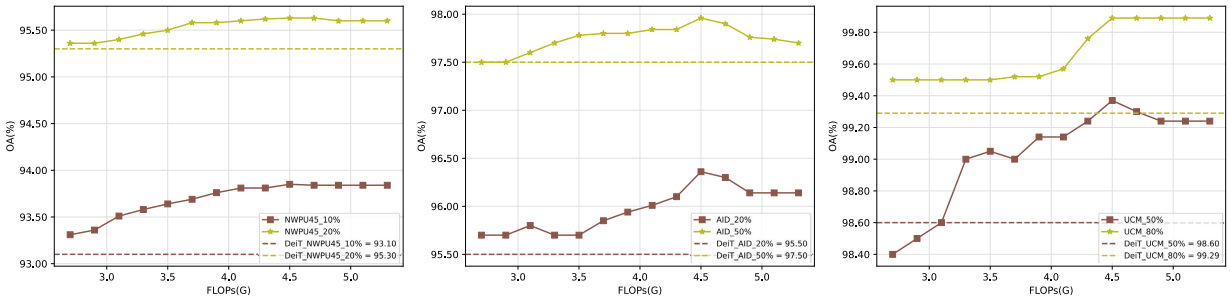
As shown in Fig. 4, SDT^{2}Net demonstrates excellent performance across six different training ratios under three datasets. The model's performance improves with increasing FLOPs, reaching a local optimum at FLOPs set to 4.5G. Therefore, in Table II, the classification accuracy of our SDT^{2}Net at each training ratio is based on the optimal training results at FLOPs $= 4.5G$. Higher accuracy implies the need for a higher computational complexity. However, as FLOPs decreases, the model's performance tends to stabilize or even slightly decrease, as shown in Fig. 4. This indicates that the advantage of SDT^{2}Net is more excellent under relatively low FLOPs constraints, suggesting its ability to provide the most suitable compression rate. SDT^{2}Net also benefits from unified token compression, indicating its capability to preserve more information by combining the advantages of ToMe and pruning.

From Fig. 4, except for the AID dataset and UC Merced dataset with a training ratio of 50%, the original DeiT's classification accuracy is lower than that of any FLOPs. In Fig. 4(b), while FLOPs are below 3.3G, DeiT's OA is relatively close to SDT^{2}Net. However, for other FLOPs, SDT^{2}Net consistently outperforms DeiT. Based on Fig. 4(c), while FLOPs are below 3.1G, DeiT surpasses SDT^{2}Net in OA. Nevertheless, with both models achieving classification accuracies above 98%, the performance gap between them is only 0.2%. These experimental results validate the superior efficiency of our SDT^{2}Net as a transformer model suitable for remote sensing scene classification.

E. Ablation Study

In this section, an ablation study is given to evaluate the contribution of each module in SDT^{2}Net on three datasets.

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(a) (b) (c)

Fig. 4. Model performance of SDT^{2}Net and DeiT under different FLOPs (G) settings on three datasets. (a) Training ratios of NWPU-RESISC45 dataset are 10% and 20%. (b) Training ratios of the AID dataset are 20% and 50%. (c) Training ratios of the UC Merced dataset are 50% and 80%.

TABLE VI
OA (%) OF SDT^{2}NET USING DIFFERENT COMPONENTS

Model	EAB	FTF	NWPU-RESISC45		AID		UC Merced	
			10%	20%	20%	50%	50%	80%
A0			93.10 ± 0.07	95.30 ± 0.07	95.50 ± 0.10	97.50 ± 0.07	98.60 ± 0.08	99.29 ± 0.12
A1	✓		93.37 ± 0.10	95.40 ± 0.04	95.80 ± 0.11	97.66 ± 0.08	98.95 ± 0.13	99.30 ± 0.12
A2		✓	93.50 ± 0.09	95.50 ± 0.07	96.03 ± 0.07	97.73 ± 0.05	99.14 ± 0.16	99.50 ± 0.08
A3	✓	✓	93.71 ± 0.12	95.59 ± 0.06	96.19 ± 0.10	97.88 ± 0.11	99.33 ± 0.14	99.76 ± 0.12
B0			93.29 ± 0.02	95.47 ± 0.01	95.80 ± 0.04	97.64 ± 0.03	98.70 ± 0.01	99.30 ± 0.01
B1	✓		93.45 ± 0.03	95.53 ± 0.02	96.00 ± 0.05	97.72 ± 0.05	99.00 ± 0.01	99.57 ± 0.01
B2		✓	93.70 ± 0.11	95.60 ± 0.02	96.21 ± 0.03	97.84 ± 0.01	99.24 ± 0.01	99.77 ± 0.02
B3	✓	✓	93.85 ± 0.02	95.63 ± 0.01	96.36 ± 0.04	97.96 ± 0.03	99.37 ± 0.05	99.89 ± 0.01

Table VI reports the efficacy of the models under three designed modules (EAB, DTC, and FTF). Group A represents the absence of the DTC module, while Group B represents the presence of the DTC module. A0 represents the original DeiT, and B3 represents the SDT^{2}Net proposed in this article.

From the results, one can see that both Groups A and B could achieve performance improvement in classification accuracy after embedding EAB and FTF modules, respectively. Specifically, the A1 model embedded with EAB achieves an accuracy improvement of 0.27%, 0.30%, and 0.35% compared to A0 at low training ratios on three datasets, respectively. B1 model improves by 0.16%, 0.20%, and 0.30% compared to B0. The A2 model, which only embeds FTF, achieves an accuracy improvement of 0.40%, 0.53%, and 0.54% compared to A0 at low training ratios on three datasets. B2 model improves by 0.41%, 0.41%, and 0.54% compared to B0. These results also indicate that embedding the FTF module has a slightly greater influence on improving model performance than embedding the EAB module. While embedding EAB and FTF simultaneously, A3 achieves the best classification results in Group A models of 93.71%, 96.19%, and 99.33%. At present, after embedding the token compression module, the classification accuracy of the B3 group model is 93.85%, 96.36%, and 99.37%, respectively. This indicates that the embedding of the DTC module can further improve model performance while reducing computational complexity.

Obviously, with the removal of designed modules, the performance of all modified models is not as good as the B3 group model, namely, SDT^{2}Net. In summary, these designed modules could make a significant contribution to the performance improvement of SDT^{2}Net.

F. Confusion Matrix

Fig. 5(a), (c), and (e) presents the confusion matrices for scene classification by SDT^{2}Net on UC Merced, AID, and NWPU-RESISC45 datasets, respectively, with training ratios of 50% for UC Merced, 20% for AID, and 10% for NWPU-RESISC45. To further highlight the superior performance of the SDT^{2}Net model, Fig. 5(b), (d), and (f) displays the confusion matrices for DeiT across the same datasets and training ratios.

Fig. 5(a) shows the confusion matrix of SDT^{2}Net on a UCM dataset with a 50% training set ratio. Except for the "denseResidential" scene, SDT^{2}Net achieves a classification accuracy of over 98% for all other scene categories. As shown in Fig. 5(c), SDT^{2}Net obtains a classification accuracy of over 86% for all scene categories on AID dataset with a 20% training set ratio. The scene classification accuracy of "airport," "desert," "meadow," "mountain," "pond," "river," "sparse residential," "storage tanks," and "viaduct" reaches 100%. The scene categories with classification accuracy below 90% are "rescue" and "square," with OA of 86% and 87%, respectively. Based on the confusion matrix of the NWPU-RESISC45 dataset with a 10% training ratio shown in Fig. 5(a), SDT^{2}Net achieves a classification accuracy of over 90% for most scene categories, especially for "chaparral," "circular farmland," "golf course," "harbor," "sea ice," and "snowberg" with a classification accuracy of 99%. The accuracy of only nine scene categories ranges from 86% to 90%, with classification accuracies of 87%, 89%, and 88% for "church," "commercial area," and "dense residential," respectively. Only the accuracy of "palace" scenario is below 80%, and the classification accuracy is 72%. In the confusion matrix of the comparison

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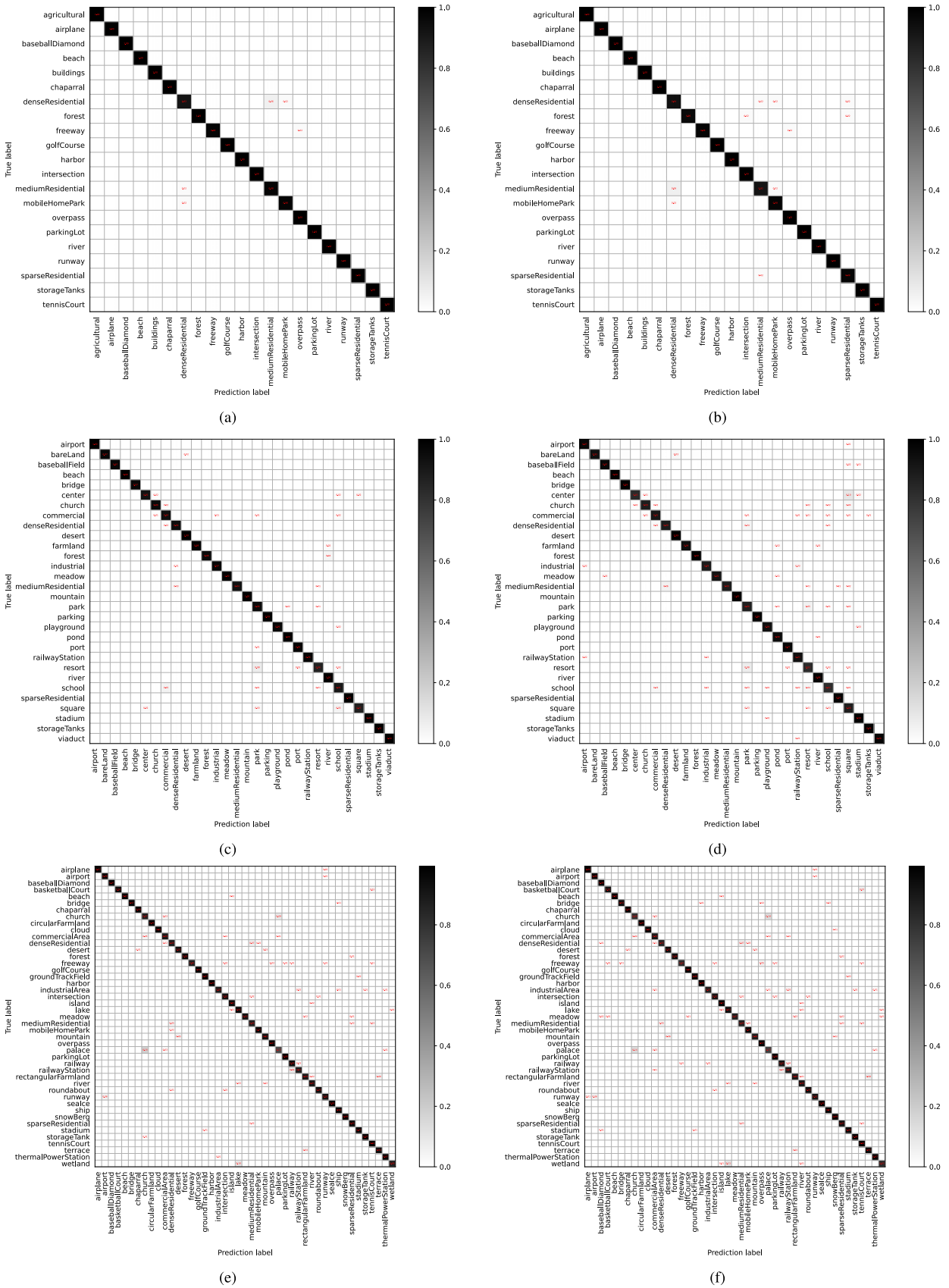
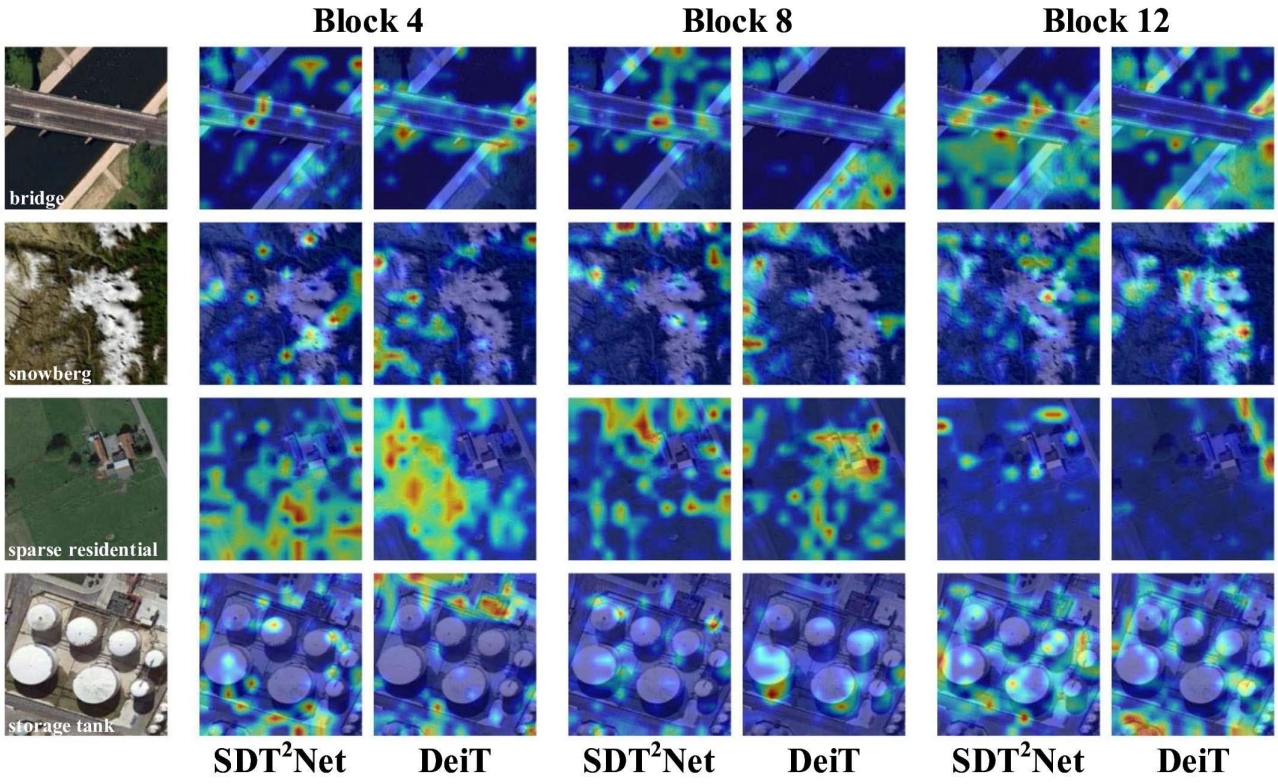
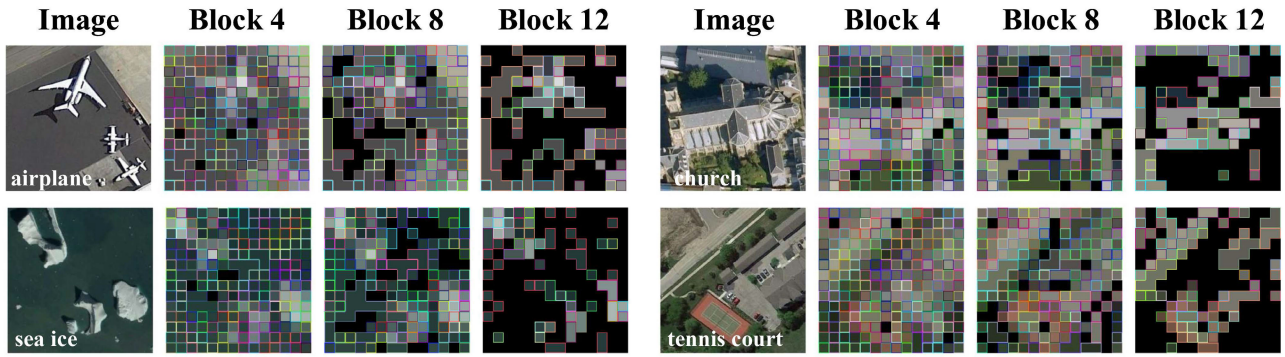


Fig. 5. Confusion matrices of SDT^{2}Net and DeiT under different training ratios on UC Merced, AID, and NWPU-RESISC45 datasets. (a) Confusion matrix of SDT^{2}Net for scene classification on the UC Merced dataset with a training ratio of 50%. (b) Confusion matrix of DeiT for scene classification on the UC Merced dataset with a training ratio of 50%. (c) Confusion matrix of SDT^{2}Net for scene classification on the AID dataset with a training ratio of 20%. (d) Confusion matrix of DeiT for scene classification on the AID dataset with a training ratio of 20%. (e) Confusion matrix of SDT^{2}Net for scene classification on the NWPU-RESISC45 dataset with a training ratio of 10%. (f) Confusion matrix of DeiT for scene classification on the NWPU-RESISC45 dataset with a training ratio of 10%.

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(a)



(b)

Fig. 6. Visualization. (a) Visualizations of images on NWPU-RESISC45 validation dataset based on SDT^{2}Net and DeiT by GradCAM method. The images are selected from NWPU-RESISC45 validation set with a training ratio of 10%, and the scene categories are "bridge," "snowberg," "sparse residential," and "storage tank." (b) Visualization of tokens compression. The images are selected from NWPU-RESISC45 validation dataset with a training ratio of 10%, and the scene categories are "airplane," "church," "sea ice," and "tennis court."

algorithm DeiT, one can be seen that SDT^{2}Net has better classification performance than DeiT on all three datasets. For example, DeiT shows significant differences on the NWPU-RESISC45 dataset compared to SDT^{2}Net in the "church," "freeway," and "meadow" scenarios, with OA being 7%, 5%, and 5% lower, respectively.

In summary, SDT^{2}Net demonstrates superior classification performance in remote sensing image scene classification. However, there is still room for improvement in the classification accuracy of easily confused scene categories. For instance, the classification accuracy of "palace" in Fig. 5(a) is only 72%, showing no improvement compared to DeiT. Therefore, embedding structural features and enhancing the backbone to effectively learn feature representations of complex scene images will be the focus of future work.

G. Visualization

To visualize the model in scene classification, we initially train our SDT^{2}Net and DeiT on the NWPU-RESISC45 dataset, and then select features of Blocks 4, 8, and 12. GradCAM [66] is utilized to obtain the class activation map for each input image. Therefore, we can visualize the most important region for final classification based on gradient information, which is the region of interest to the network. As illustrated in Fig. 6(a), we randomly select four scenes from the test set of the NWPU-RESISC45 dataset with a training ratio of 10% for visualization, namely, "bridge," "snowberg," "sparse residential," and "storage tank."

From the visualization results, as the number of blocks increases, SDT^{2}Net focuses more accurately on the target area compared to DeiT. For example, SDT^{2}Net focuses more on the bridge body and water near the bridge, while DeiT focuses on

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the nonbridge part in Block 12 of the bridge scene. DeiT is more likely to confuse "bridge" with "overpass" and "ship," resulting in misclassification. This can be verified in Fig. 5(f), where the classification accuracy of the "bridge" category in the confusion matrix of DeiT is only 93%, which is 4% lower than SDT^{2}Net. There are probabilities of 1%, 2%, and 2%, respectively, to misclassify "bridge" into "harbor," "overpass," and "ship" categories. Compared to DeiT, SDT^{2}Net achieves higher classification accuracy with fewer tokens, such as the scenes of "snowberg" and "sparse residential." While facing scenarios that contain multiple objects such as "storage tank," although SDT^{2}Net and DeiT focus on different regions in the heat maps of Blocks 4 and 8, both recognize multiple storage tanks in final Block 12, and SDT^{2}Net pay much attention to the main components of storage tanks.

As tokens are compressed layer by layer, the number of effective tokens decreases. In order to more intuitively demonstrate the token compression strategy by SDT^{2}Net, Fig. 6(b) visually illustrates the number of valid tokens remaining in the network and their positions in the original image under different blocks, and the pruned tokens are replaced with a black background. It is evident that throughout the layer-by-layer compression stage, the number of compressed tokens increases. Crucial tokens that help distinguish scene categories are retained and clustered around the target object. For the "sea ice" scene, the remaining tokens of the last block are gathered near "ice," while the tokens with the "ocean" background part are pruned. Simultaneously, the adjacent and merged tokens are presented with the same color block and border. For the "airplane," "church," and "tennis court" scenarios, as the number of network layers grows, more and more remaining tokens are gradually merged with surrounding tokens. Especially, one can be seen from the "church" scene, the main building is composed of many homogeneous tokens, which helps to reduce network parameters without affecting the classification accuracy.

In summary, the proposed SDT^{2}Net could pay much attention to important discriminative semantic regions in features, while regions with less information are weakly activated.

V. Conclusion

For constructing a distinguishable transformer network that can adaptively select tokens for efficiently remote sensing scene classification, this article proposes an SDT^{2}Net, adaptively performing token pruning and ToMe in a differentiable manner under a given FLOPs constraint. Considering the adaptive selection and compression of discriminative features, we design an EAB and DTC mechanism, which formalizes the token compression rate as a style embedded in the network, thereby transforming it into an optimization problem for parameter learning. Furthermore, an FTF module is developed via the matrix Taylor polynomial method, which employs a structured normalized fast global covariance pooling style to efficiently integrate classification tokens and high-order visual tokens. Our experimental results demonstrate that SDT^{2}Net is superior to previous state-of-the-art remote sensing scene classification methods. Although SDT^{2}Net achieves excellent classification performance in remote sensing image scene classification, there still exist misclassifications in a small portion of complex scene categorizations.

The structural information of various regions or targets within a complex background can better distinguish the importance of different regions in scene images and capture the global and local features of remote sensing scene images. Then, constructing a more effective way for learning scene features by integrating structural information (such as graph convolution network) among scene targets, while considering the token pruning and ToMe strategies within the transformer network, is one of the future works. Furthermore, the adaptive learning of the MHSA mechanism in the transformer can effectively capture important feature information of remote sensing scene images. Therefore, one of the future works is to rationally combine MHSA with a token compression strategy and design a lightweight network structure suitable for remote sensing image scene classification.

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